

Chapter 2, Section 3

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1. Show that a morphism $f : X \rightarrow Y$ is locally of finite type if and only if for every open affine subset $V = \operatorname{Spec} B$ of Y , $f^{-1}(V)$ can be covered by open affine subsets $U_j = \operatorname{Spec} A_j$, where each A_j is a finitely generated B -algebra.

Proof. We reduce to proving the following statement: let $f : X \rightarrow Y$ be a morphism of schemes with $Y = \operatorname{Spec} B$, which can be covered by open affine subsets $V_i = \operatorname{Spec} B_i$, such that for each i , $f^{-1}(V_i)$ can be covered by open affine subsets $U_{ij} = \operatorname{Spec} A_{ij}$, where each A_{ij} is a finitely generated B_i -algebra. Then X can be covered by open affine subsets $U_k = \operatorname{Spec} A_k$, where each A_k is a finitely generated B -algebra.

We show each B_i can be chosen to be a finitely generated B -algebra, then by restriction of scalars each A_{ij} is a finitely generated B -algebra, which proves the statement. For each i , there exists $g_i \in B$ such that $D(g_i) \subseteq V_i$, where $D(g_i) \cong \operatorname{Spec} B_{g_i}$. Then $f^{-1}(D(g_i)) \subseteq f^{-1}(V_i)$, so we can cover $f^{-1}(D(g_i))$ by open affines of the form $\operatorname{Spec}((A_{ij})_{h_k})$ with $h_k \in A_{ij}$. Since each $(A_{ij})_{h_k}$ is a finitely generated A_{ij} -algebra, and each $(A_{ij})_{h_k}$ is a finitely generated $B_{g_i} \cong (B_i)_{\bar{g}_i}$, where each $\bar{g}_i$ is the image of g_i in B_i , what we have shown is each $f^{-1}(D(g_i))$ can be covered by open affine subsets $U'_{ij} = \operatorname{Spec} A'_{ij}$, where each A'_{ij} is a finitely generated B_{g_i} -algebra. Each B_{g_i} is a finitely generated B -algebra, generated by 1 and $1/g_i$, which is what we wanted to show. $\square$

2. A morphism $f : X \rightarrow Y$ of schemes is *quasi-compact* if there is a cover of Y by open affines V_i such that $f^{-1}(V_i)$ is quasi-compact for each i . Show that f is quasi-compact if and only if for *every* open affine subset $V \subseteq Y$, $f^{-1}(V)$ is quasi-compact.

Proof. We reduce to proving the following statement: let $f : X \rightarrow Y$ be a morphism of schemes with Y affine, which can be covered by open subsets V_i such that $f^{-1}(V_i)$ is quasi-compact in X . Then X is quasi-compact. Indeed, Y is quasi-compact, so a finite number of i will do, and a finite union of quasi-compact sets is quasi-compact, and $X = f^{-1}(Y) = f^{-1}(\bigcup_{i=1}^n V_i) = \bigcup_{i=1}^n f^{-1}(V_i)$, hence X is quasi-compact. $\square$

3. (a) Show that a morphism $f : X \rightarrow Y$ is of finite type if and only if it is locally of finite type and quasi-compact.
 (b) Conclude from this that f is of finite type if and only if for *every* open affine subset $V = \operatorname{Spec} B$ of Y , $f^{-1}(V)$ can be covered by a finite number of open affines $U_j = \operatorname{Spec} A_j$, where each A_j is a finitely generated B -algebra.
 (c) Show also if f is of finite type, then for *every* open affine subset $V = \operatorname{Spec} B \subseteq Y$, and for *every* open affine subset $U = \operatorname{Spec} A \subseteq f^{-1}(V)$, A is a finitely generated B -algebra.

Proof.

- (a) Suppose $f : X \rightarrow Y$ is of finite type. Then by definition it is also locally of finite type, so there exists a covering of Y by open affine subsets $V_i = \operatorname{Spec} B_i$, such that each i , $f^{-1}(V_i)$ can be covered by a finite number of open affine subsets $U_{ij} = \operatorname{Spec} A_{ij}$, where each A_{ij} is a finitely generated B_i -algebra. Each $f^{-1}(V_i)$ is quasi-compact since every affine scheme is quasi-compact and $f^{-1}(V_i)$ is a finite union of quasi-compact sets. Conversely, if $f : X \rightarrow Y$ is locally of finite type and quasi-compact, then each $f^{-1}(V_i)$ is quasi-compact, so it can be covered by a finite number of open affines $U_{ij} = \operatorname{Spec} A_{ij}$, where each A_{ij} is a finitely generated B_i -algebra.

- (b) A morphism $f : X \rightarrow Y$ is of finite type if and only if it is locally of finite type and quasi-compact if and only if for every open affine subset $V = \operatorname{Spec} B$ of Y , $f^{-1}(V)$ is quasi-compact and can be covered by open affines $U_j = \operatorname{Spec} A_j$, where each A_j is a finitely generated B -algebra if and only if for every open affine subset $V = \operatorname{Spec} B$ of Y , $f^{-1}(V)$ can be covered by a finite number of open affines $U_j = \operatorname{Spec} A_j$, where each A_j is a finitely generated B -algebra.
- (c) By restricting the domain of f , it suffices to prove the following: if X is a scheme over a ring B that can be covered by a finite number of open subsets that are the spectra of finitely generated B -algebras, then for every open affine $U = \operatorname{Spec} A \subseteq X$, A is a finitely generated B -algebra. By Nike's trick, we can cover U by the distinguished basic sets $\operatorname{Spec} A_f$, where each A_f is a finitely generated B -algebra. Since U is quasi-compact, a finite number will do.

So now, we have reduced to the following purely algebraic problem: A is a B -algebra, $f_1, \dots, f_r$ are a finite number of elements of A , which generate the unit ideal, and each localization A_{f_i} is a finitely generated B -algebra. Then A is finitely generated over B . Write $\sum_{i=1}^r g_i f_i$. Let S be the union of $\{g_1, \dots, g_r, f_1, \dots, f_r\}$ and a finite subset of A such that the image of it under the natural map $A \rightarrow A_{f_i}$ generates A_{f_i} as a B -algebra for all i , and let $B[S]$ be the B -subalgebra of A generated by S . Localization is left-exact, so $B[S]_{f_i} \rightarrow A_{f_i}$ is injective for all i , and it is surjective as well by construction of S . Thus, the inclusion $B[S] \rightarrow A$ viewed as a $B[S]$ -module homomorphism is an isomorphism by the following statement from commutative algebra: let R be any ring, let M be any R -module, and let $h_1, \dots, h_n$ be elements of R that generate the unit ideal. If $M_{h_i} = 0$ for all i , then $M = 0$. Indeed, if $\mathfrak{p}$ is any prime ideal of R , then $h_i \notin \mathfrak{p}$ for some i . Denote $\bar{h}_i$ the image of h_i in $A_{\mathfrak{p}}$. Then, $M_{\mathfrak{p}} = (M_{\mathfrak{p}})_{\bar{h}_i} = (M_{h_i})_{\mathfrak{p}R_{h_i}} = 0$ (A.M. §3).

□

Theorem 0.1 (Nike's Trick). *Let X be a scheme, and let $U_i = \operatorname{Spec} A_i$, $i = 1, 2$, be open affine subsets of X . Then there is an open cover of $U_1 \cap U_2$ consisting of open sets that are distinguished basic open sets in both U_i .*

Lemma 0.2. *If $(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ is a morphism of locally ringed spaces, then $f^{-1}(Y_g) = X_{f^\#(g)}$ for any $g \in \Gamma(Y, \mathcal{O}_Y)$.*

Proof. Pick $g \in A_1$ such that $\operatorname{Spec} (A_1)_g = (U_1)_g \subseteq U_1 \cap U_2$, and set $U_1 = (U_1)_g$, so we have an open immersion $\iota : U_1 \hookrightarrow U_2$, which induces a ring homomorphism $\varphi : A_2 \rightarrow (A_1)_g$. Pick $h \in A_2$ such that $(U_2)_h \subseteq U_1$. By lemma 1, $\iota^{-1}((U_2)_h) = (U_1)_{\varphi(h)}$, and since ι is an open immersion, $(U_1)_{\varphi(h)} \cong (U_2)_h$ (we can assume $\varphi(h) \in A$). □

4. Show that a morphism $f : X \rightarrow Y$ is finite if and only if for every open affine subset $V = \operatorname{Spec} B$ of Y , $f^{-1}(V)$ is affine, equal to $\operatorname{Spec} A$, where A is a finite B -module.

Proof. Let $f : X \rightarrow Y$ be a finite morphism of schemes. Then there exists an open covering of Y by sets $V_i = \operatorname{Spec} B_i$, such that for each i , $f^{-1}(V_i)$ is affine, equal to $\operatorname{Spec} A_i$, where A_i is a finite B_i -module. Let $V = \operatorname{Spec} B$ be an open subset of Y . By Nike's trick, we can cover V by open affines that are distinguished open set in V and some V_i , i.e., open sets of the form $\operatorname{Spec} B_g = \operatorname{Spec} (B_i)_h$ for some $g \in B, h \in B_i$. Since V is quasi-compact, a finite number will do. By lemma 1, $f^{-1}(\operatorname{Spec} (B_i)_h) = \operatorname{Spec} (A_i)_h \subseteq f^{-1}(V)$. Since $(A_i)_h$ is a finite $(B_i)_h$ -module and $B_g \cong (B_i)_h$, we deduce that there exists a finite cover of V by basic open sets $\operatorname{Spec} B_{g_j}$ for some $g_j \in B$ such that $f^{-1}(\operatorname{Spec} B_{g_j}) = \operatorname{Spec} C_j$, where each C_j is a finite B_{g_j} -module. We show $X' = f^{-1}(V)$ is affine using the criterion in (Ex. 2.17b). The restriction of f to $X' \rightarrow \operatorname{Spec} B$ induces a ring homomorphism $B \rightarrow A = \Gamma(X', \mathcal{O}_X)$. Denote $\bar{g}_i$ the image of g_i in A . Since g_i generate the unit ideal in B , its image in A also generate the unit ideal. Also, by lemma 1 $X'_{\bar{g}_j} = f^{-1}(\operatorname{Spec} B_{g_j}) = \operatorname{Spec} C_j$, hence $X' = \operatorname{Spec} A$. It remains to show A is a finite B -module, which we reduce to the following algebraic problem: let A be a B -algebra, let $g_i \in B$ ($1 \leq i \leq n$) generate the unit ideal, and suppose A_{g_i} is a finite B_{g_i} -module. Then A is a finite B -module. The proof is identical to (Ex. 3.3c), so we omit this part. □

6. Let X be an integral scheme. Show that the local ring $\mathcal{O}_\xi$ of the generic point ξ of X is a field. It is called the *function field* of X , and is denoted by $K(X)$. Show also that if $U = \operatorname{Spec} A$ is any open affine subset of X , then $K(X)$ is isomorphic to the quotient field of A .

Proof. Any nonempty open set U of X must contain ξ since $X - U$ is a proper closed subset of X . In particular, any element of $\mathcal{O}_\xi$ can be represented as a pair $(\text{Spec } A, f)$ where $\text{Spec } A$ is an open affine set in X and $f \in A$. We further assume $\text{Spec } A$ is connected, so A is an integral domain. If f is a nonzero element of A , then $(\text{Spec } A, f)$ is a nonzero element of $\mathcal{O}_\xi$ with inverse $(\text{Spec } A, f)^{-1} = (\text{Spec } A_f, 1/f)$, hence $\mathcal{O}_\xi$ is a field. Lastly, the distinguished open sets $\text{Spec } A_f$ form a neighborhood basis of ξ , so any element of $\mathcal{O}_\xi$ can be written as $(\text{Spec } A_f, a/f^n)$, which is the quotient field of A . $\square$

8. *Normalization.* A scheme is *normal* if all of its local rings are integrally closed domains. Let X be an integral scheme. For each open affine subset $U = \text{Spec } A$ of X , let $\tilde{A}$ be the integral closure of A in its quotient field, and let $\tilde{U} = \text{Spec } \tilde{A}$. Show that one can glue the schemes $\tilde{U}$ to obtain a normal integral scheme $\tilde{X}$, called the *normalization* of X . Show also that there is a morphism $\tilde{X} \rightarrow X$, having the following universal property: for every normal integral scheme Z , and for every dominant morphisms $f : Z \rightarrow X$, f factors uniquely through $\tilde{X}$. If X is of finite type over a field k , then the morphism $\tilde{X} \rightarrow X$ is a finite morphism.

Lemma 0.3. *If $f : Z \rightarrow X$ is a dominant morphism of schemes with Z reduced, then $f^\# : \mathcal{O}_X \rightarrow f_*\mathcal{O}_Z$ is injective.*

Proof. Let U be any open subset of X . We want to show if $g \in \Gamma(U, \mathcal{O}_X)$ such that $f^\#(g) = 0 \in \Gamma(f^{-1}(U), \mathcal{O}_Z)$, then $g = 0$. By lemma 1, we have $f^{-1}(U_g) = (f^{-1}(U))_{f^\#(g)}$, and if $f^\#(g) = 0$, then $(f^{-1}(U))_{f^\#(g)} = \emptyset$, so U_g must not meet the $f(Z)$. But $f(Z)$ is dense in X , and U_g is an open set in X by (Ex. 2.16a), so $U_g = \emptyset$, which implies g is nilpotent (this result corresponds to the algebraic fact that the intersection of all prime ideals of a ring is the nilradical of the ring), hence $g = 0$. $\square$

Proof. Let $X = \bigcup \text{Spec } A_i$ be an open affine covering of X where each A_i is an integral domain. For each $i \neq j$, we have an identification $\varphi_{ij} : U_i \rightarrow U_j$, which is an isomorphism of open subschemes $U_i \subseteq \text{Spec } A_i, U_j \subseteq \text{Spec } A_j$. By Nike's trick, there exists an open covering of U_i by basic open sets $\text{Spec } (A_i)_{f_k}$ with $f_k \in A_i$ such that $\varphi_{ij}(\text{Spec } (A_i)_{f_k}) = \text{Spec } (A_j)_{g_k}$ with $g_k \in A_j$. For each i , let $\tilde{A}_i$ be the integral closure of A_i in its quotient field (note by (Ex. 6), every A_i has the same quotient field), and let $\pi_i : \text{Spec } \tilde{A}_i \rightarrow \text{Spec } A_i$ be the morphism induced by the inclusion $A_i \hookrightarrow \tilde{A}_i$. We have $\pi_i^{-1}(\text{Spec } (A_i)_{f_k}) = \text{Spec } (\tilde{A}_i)_{f_k}$ and $(\tilde{A}_i)_{f_k} \cong (\tilde{A}_j)_{g_k}$, so we can naturally glue open subsets of X' that are of the form $\pi_i^{-1}(U_i)$ using φ_{ij} to obtain $\tilde{X}$. The morphism $\pi : \tilde{X} \rightarrow X$ is obtained by glueing π_i accordingly.

Now suppose Z is a normal integral scheme, and let $f : Z \rightarrow X$ be a dominant morphism of schemes. It is clear we can assume $X = \text{Spec } A$, where A is an integral domain. Then f induces a ring homomorphism $\varphi : A \rightarrow B = \Gamma(Z, \mathcal{O}_Z)$. Let $\tilde{X} = \text{Spec } \tilde{A}$ be the normalization of X with associated morphism $\pi : \tilde{X} \rightarrow X$ induced by the inclusion homomorphism $\iota : A \rightarrow \tilde{A}$, where $\tilde{A}$ is the integral closure of A in its quotient field. We want to show there exists a unique morphism $\tilde{f} : Z \rightarrow \tilde{X}$ such that $f = \pi \circ \tilde{f}$. Being integrally closed is a local property (A.M. 5.13), so B is integrally closed (it is automatically an integral domain by definition of an integral scheme). Also, φ is injective by lemma 2. Since $\tilde{X}$ is affine, by the bijection in (Ex. 24), we have reduced to proving the following universal property for the integral closure of a domain $A \xrightarrow{\iota} \tilde{A}$: for any injective homomorphism $\varphi : A \rightarrow B$ where B is an integrally closed domain, there exists a unique homomorphism $\psi : \tilde{A} \rightarrow B$ such that $\varphi = \psi \circ \iota$. Any injective homomorphism between integral domains induces an inclusion of fraction fields, so let $\Phi : \text{Frac}(A) \rightarrow \text{Frac}(B)$ be induced by φ , where $\Phi|_A = \varphi$. Note that we have inclusions $A \subseteq \tilde{A} \subseteq \text{Frac}(A)$, thus we claim $\psi = \Phi|_{\tilde{A}} : \tilde{A} \rightarrow \text{Frac}(B)$ is the desired ring homomorphism. It suffices to show the image of ψ is contained in B . If $f \in \tilde{A}$, then there exists an equation of integral dependence $f^n + a_1 f^{n-1} + \dots + a_n = 0$ where $a_i \in A$. Then $\Phi(f)$ has an equation of integral dependence $\Phi(f)^n + \varphi(a_1)\Phi(f)^{n-1} + \dots + \varphi(a_n) = 0$, and since B is integrally closed, $\Phi(f)$ must be an element of B . Since any other $\psi' : \tilde{A} \rightarrow B$ such that $\varphi = \psi' \circ \iota$ must agree with Φ on $\tilde{A}$, ψ is unique by construction. $\square$

10. Fibers of a Morphism.

- (a) If $f : X \rightarrow Y$ is a morphism, and $y \in Y$ a point, show that $\text{sp}(X_y)$ is homeomorphic to $f^{-1}(y)$ with the induced topology.
- (b) Let $X = \text{Spec } k[s, t]/(s - t^2)$, let $Y = \text{Spec } k[s]$, and let $f : X \rightarrow Y$ be the morphism defined by sending $s \rightarrow s$. If $y \in Y$ is the point $a \in k$ with $a \neq 0$, show that the fiber X_y , consists of two points, with residue

field k . If $y \in Y$ corresponds to $0 \in k$, show that the fiber X_y is a nonreduced one-point scheme. If η is the generic point of Y , show that X_η is a one-point scheme, whose residue field is an extension of degree two of the residue field of η . (Assume k is algebraically closed.)

Proof.

- (a) We can assume Y to be affine by taking any open affine neighborhood of $y \in Y$ and restricting f to its preimage in X . Also, if $U_i = \text{Spec } A_i$ is an open affine cover of X , then $f^{-1}(y) = \bigcup_i (\text{Spec } A_i \cap f^{-1}(y))$, and $\text{sp}(X_y) = \bigcup_i \text{sp}(U_i \times_Y k(y))$, so we can also assume X to be affine. Let $X = \text{Spec } A$ and $Y = \text{Spec } B$, then f induces a ring homomorphism $f^\# : B \rightarrow A$. A point $y \in Y$ corresponds to a prime ideal $\mathfrak{q}$ in B , where $k(y) = B_{\mathfrak{q}}/\mathfrak{q}B_{\mathfrak{q}}$. Thus, $f^{-1}(y)$ is the set of all prime ideals $\mathfrak{p}$ in A such that $f^{\#-1}(\mathfrak{p}) = \mathfrak{q}$. Next, we look at $\text{sp}(X_y)$. Notice that $X_y = \text{Spec } A \otimes_B k(y)$, and $A \otimes_B k(y) = A \otimes_B B_{\mathfrak{q}}/\mathfrak{q}B_{\mathfrak{q}} = A_{\mathfrak{q}}/\mathfrak{q}A_{\mathfrak{q}}$. The prime ideals of $A_{\mathfrak{q}}/\mathfrak{q}A_{\mathfrak{q}}$ correspond to prime ideals of A that contain the image of $\mathfrak{q}$ and does not contain meet the image of $B - \mathfrak{q}$, which are precisely the prime ideals of A such that $f^{\#-1}(\mathfrak{p}) = \mathfrak{q}$.
- (b) $X_y = \text{Spec } k[t]/(a - t^2)$, $X_\eta = \text{Spec } k(s)[\sqrt{s}]$. Any element of $k(s)[\sqrt{s}]$ can be written as $F + G\sqrt{s}$, where $F, G \in k(s)$. It is a field, since $(F + G\sqrt{s})^{-1} = (F - G\sqrt{s})/(F^2 - sG^2)$.

□

11. Closed Subschemes.

- (a) Closed immersions are stable under base extension: if $f : Y \rightarrow X$ is a closed immersion, and if $X' \rightarrow X$ is any morphism, then $f' : Y \times_X X' \rightarrow X'$ is also a closed immersion.
- (b) If Y is a closed subscheme of an affine scheme $X = \text{Spec } A$, then Y is also affine, and in fact Y is the closed subscheme determined by a suitable ideal $\mathfrak{a} \subseteq A$ as the image of the closed immersion $\text{Spec } A/\mathfrak{a} \rightarrow \text{Spec } A$.
- (c) Let Y be a closed subset of a scheme X , and give Y the reduced induced subscheme structure. If Y' is any other closed subscheme of X with the same underlying topological space, show that the closed immersion $Y \rightarrow X$ factors through Y' . We express this property by saying that the reduced induced structure is the smallest subscheme structure on a closed subset.
- (d) Let $f : Z \rightarrow X$ be a morphism. Then there is a unique closed subscheme Y of X with the following property: the morphism f factors through Y , and if Y' is any other closed subscheme of X through which f factors, then $Y \rightarrow X$ factors through Y' also. We call Y the *scheme-theoretic image* of f . If Z is a reduced scheme, then Y is just the reduced induced structure on the closure of the image $f(Z)$.

Proof.

- (a) Consider the special case when $X = \text{Spec } A$, $Y = \text{Spec } A/\mathfrak{a}$, and $X' = \text{Spec } B$ where $\mathfrak{a}$ is an ideal of A and B is any A -algebra. The natural map $A \rightarrow A/\mathfrak{a}$ induces a closed immersion $Y \rightarrow X$, and the structure homomorphism $A \rightarrow B$ induces a morphism of schemes $X' \rightarrow X$. The fiber product $Y \times_X X'$ is equal to the spectra of the tensor product $A/\mathfrak{a} \otimes_A B$, which is isomorphic to $B/\mathfrak{a}B$. The induced structure morphism $Y \times_X X' \rightarrow X'$ corresponds to the canonical homomorphism $B \rightarrow B/\mathfrak{a}B$, thus $Y \times_X X' \rightarrow X'$ is a closed immersion. In other words, this property of closed immersions corresponds to the algebraic fact that the tensor operation is a right exact functor. In the general case, we can still assume X to be affine by taking an open affine cover of X . Then Y is an affine scheme by part (b), and if U is any open subset of X' , $f'^{-1}(U) = Y \times_X U$, so by taking an open affine cover of X' , we can reduce to the case when X' is affine, which is just the special case as in above.
- (b) Let Y be a closed subscheme of an affine scheme $X = \text{Spec } A$, and let $\varphi : A \rightarrow B = \Gamma(Y, \mathcal{O}_Y)$ be the induced ring homomorphism of global sections. Fix $y \in Y$, let $V = \text{Spec } C$ be an open affine neighborhood of y as a subspace of Y , and let $\rho : B \rightarrow C$ be the restriction homomorphism. By definition of the subspace topology, there exists an open set U of X such that $V = U \cap Y$. We can cover U by distinguished open sets, and at least one of them must contain y , so let X_f be such set with $f \in A$. By lemma 1, $X_f \cap Y = Y_{\varphi(f)} = \text{Spec } C_{\rho(\varphi(f))}$. Thus, we can cover Y by open affines of the form $X_{f_i} \cap Y$. Since Y is homeomorphic to a closed subset of a quasi-compact set, it is also quasi-compact, so a finite number will do. By adding some more f_i with $D(f_i) \cap Y = \emptyset$ by taking an open cover of $X - Y$, we assume X_{f_i} cover X . Such collection of X_{f_i} cover X if and only if f_i generate the unit ideal in A , so $\varphi(f_i)$ must generate the unit ideal of B . Hence, Y is affine by (Ex. 2.17b). The quotient ring $A/\ker \varphi$ is a subring of B , so the affine scheme $X' = \text{Spec } A/\ker \varphi$ contains Y as a dense subset. X' is also homeomorphic to a closed subset of X , hence $Y = \text{Spec } A/\ker \varphi$.

- (c) By taking an affine cover of X , we reduce to the case when $X = \operatorname{Spec} A$ is affine. Let $\mathfrak{a}$ be the ideal in A that corresponds to the reduced induced structure of Y . By part (b), there exists an ideal $\mathfrak{b}$ in A such that $Y' = \operatorname{Spec} A/\mathfrak{b}$. A morphism of schemes $Y \rightarrow Y'$ corresponds to a ring homomorphism $A/\mathfrak{b} \rightarrow A/\mathfrak{a}$. Recall that $\mathfrak{a}$ is the largest ideal of A such that $V(\mathfrak{a}) = \operatorname{sp}(Y) = \operatorname{sp}(Y')$; in particular, $\mathfrak{b} \subseteq \mathfrak{a}$, so the natural projection map $A/\mathfrak{b} \rightarrow A/\mathfrak{a}$ is well-defined and is the desired ring homomorphism.
- (d) Again, reduce to the affine case. A morphism of affine schemes $f : \operatorname{Spec} B \rightarrow \operatorname{Spec} A$ correspond to a ring homomorphism $\varphi : A \rightarrow B$. Then $Y = \operatorname{Spec} A/\ker \varphi$ is the desired closed subscheme. If Z is reduced, then B is a reduced ring. Then $\ker \varphi$ contains the nilradical of A , so $A/\ker \varphi$ is reduced, hence the intersection of all prime ideals of $A/\ker \varphi$ is the zero ideal.

□

13. Properties of Morphisms of Finite Type.

- (a) A closed immersion is a morphism of finite type.
- (b) A quasi-compact open immersion is of finite type.
- (c) A composition of two morphisms of finite type is of finite type.
- (d) Morphisms of finite type are stable under base extension.
- (e) If X and Y are schemes of finite type over S , then $X \times_S Y$ is of finite type over S .
- (f) If $X \xrightarrow{f} Y \xrightarrow{g} Z$ are two morphisms, and if f is quasi-compact, and $g \circ f$ is of finite type, then f is of finite type.
- (g) If $f : X \rightarrow Y$ is a morphism of finite type, and if Y is noetherian, then X is noetherian.

Proof.

- (a) Let $f : Y \rightarrow X$ be a closed immersion. By abuse of notation, also denote Y as a closed subset of $\operatorname{sp}(X)$. If U is quasi-compact in X , then $f^{-1}(U) = U \cap Y$, which is quasi-compact since any closed subset of a quasi-compact set is also quasi-compact, so f is a quasi-compact morphism. Also, if $U = \operatorname{Spec} A$ is an open affine subset of X , then $f^{-1}(U) = U \cap Y$ is a closed subscheme of U equal to the spectra of $A/\mathfrak{a}$ for some ideal $\mathfrak{a}$ of A by (Ex. 3.11b), which is a finitely generated A -algebra, so f is also locally of finite type. Hence, f is of finite type.
- (b) Let $f : Y \rightarrow X$ be an open immersion. If $U = \operatorname{Spec} A$ is any open affine set in X , then $f^{-1}(U) = U \cap Y$ can be covered by distinguished open sets of U , which are spectra of A_f for some $f \in A$. Any such A_f is a finitely generated A -algebra. Hence, an open immersion is locally of finite type, so a quasi-compact open immersion is of finite type by (Ex. 3.3a).
- (c) Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be morphisms of finite type. Let $W = \operatorname{Spec} C$ be an open affine subset of Z . Since g is of finite type, $g^{-1}(W)$ can be covered by a finite number of open affines $V_i = \operatorname{Spec} B_i$, where each B_i is a finitely generated C -algebra, and similarly each $f^{-1}(V_i)$ can be covered by a finite number of open affines $U_{ij} = \operatorname{Spec} A_{ij}$, where each A_{ij} is a finitely generated B_i -algebra. Each A_{ij} is also a finitely-generated A -algebra since B_i is a finitely generated A -algebra, and there are a finite number of U_{ij} with $(g \circ f)^{-1}(W) = f^{-1}(g^{-1}(W)) = \bigcup U_{ij}$, $g \circ f$ is a morphism of finite type.
- (d) This corresponds to the algebraic fact that if A is a finitely generated B -algebra and C is any other B -algebra, then $A \otimes_B C$ is a finitely generated C -algebra.
- (e) This corresponds to the algebraic fact that the tensor product of two finitely generated algebras is also finitely generated.
- (f) By (Ex. 3.3a), it suffices to show f is locally of finite type. Let $W = \operatorname{Spec} C$ be an open affine subset of Z , and let $V_i = \operatorname{Spec} B_i$ be an open affine cover of $g^{-1}(W)$. By (Ex. 3.3c), for each i , we can cover $f^{-1}(V_i)$ by open affines $U_{ij} = \operatorname{Spec} A_{ij}$ where each A_{ij} is a finitely generated C -algebra. We have a composition of structure morphisms $C \rightarrow B_i \rightarrow A_{ij}$, where A_{ij} is finitely generated over the image of C in A_{ij} . Since the image of C in A_{ij} is contained in the image of B_i in A_{ij} , A_{ij} is also a finitely generated B_i -algebra.
- (g) This corresponds to the fact that any finitely generated algebra over a noetherian ring is also noetherian as a ring.

□

16. *Noetherian Induction.* Let X be a noetherian topological space, and let $\mathcal{P}$ be a property of closed subsets of X . Assume that for any closed subset Y of X , if $\mathcal{P}$ holds for every proper closed subset of Y , then $\mathcal{P}$ holds for Y . (In particular, $\mathcal{P}$ must hold for the empty set.) Then $\mathcal{P}$ holds for X .

Proof. Suppose $\mathcal{P}$ does not hold for X , and set $X_1 = X$. Recursively define X_{n+1} to be any proper closed subset of X_n such that $\mathcal{P}$ does not hold. This is a nonterminating decreasing sequence of closed subsets of X by construction, a contradiction. $\square$

20. *Dimension.* Let X be an integral scheme of finite type over a field k (not necessarily algebraically closed). Use appropriate results from (I, §1) to prove the following.

- (a) For any closed point $P \in X$, $\dim X = \dim \mathcal{O}_P$, where for rings, we always mean the Krull dimension.
- (b) Let $K(X)$ be the function field of X . Then $\dim X = \text{tr.d. } K(X)/k$.
- (c) If Y is a closed subset of X , then $\text{codim}(Y, X) = \inf\{\dim \mathcal{O}_{P, X} \mid P \in Y\}$.
- (d) If Y is a closed subset of X , then $\dim Y + \text{codim}(Y, X) = \dim X$.
- (e) If U is a nonempty open subset of X , then $\dim U = \dim X$.
- (f) If $k \subseteq k'$ is a field extension, then every irreducible component of $X' = X \times_k k'$ has dimension $= \dim X$.

Proof.

- (a) Let V be a variety over k . The bijection between the open sets of V and open sets of $t(V)$ induced by the map α in the proof of (2.6) implies $\dim V = \dim t(V)$. If X is an integral scheme of finite type over a field k , then we can cover X by a finite number of open affines $U_i = \text{Spec } A_i$, where each A_i is of the form $k[x_1, \dots, x_{r_i}]/\mathfrak{p}_i$ for some $r_i > 0$ and prime ideal $\mathfrak{p}_i$ in $k[x_1, \dots, x_{r_i}]$. Then $P \in U_i$ for some i , and if P is closed, then it corresponds to a maximal ideal $\mathfrak{m}_P$ in A_i , so by (A.M. 11.25) and (I, 1.7), $\dim \mathcal{O}_P = \dim A_i = \dim U_i$. Also, $\dim U_i = \dim X$ for all i by (I, Ex. 1.10b) and from the fact that $\dim U_i = \dim U_j$. Indeed, X itself is irreducible since it is an integral scheme and therefore has a unique generic point. Then $U_i \cap U_j \neq \emptyset$ for any i, j , so for all $P \in U_i \cap U_j$, $\dim U_i = \dim \mathcal{O}_{U_i, P} = \dim \mathcal{O}_{U_j, P} = \dim U_j$. Hence, $\dim X = \dim \mathcal{O}_P$ for all closed $P \in X$.
- (b) By part (a), (II, Ex. 3.6), (I, 1.8A), $\text{tr.d. } K(X)/k = \dim U_i = \dim X$.
- (c) Let $X = \text{Spec } A$, where A is an integral domain that is finitely generated over a field k , and let Y be an irreducible, closed subset of X . The closed subscheme Y corresponds to a prime ideal $\mathfrak{p}$ in A , and any irreducible closed set containing Y corresponds to a prime ideal contained in $\mathfrak{p}$, so $\text{codim } YX$ equals to the height of $\mathfrak{p}$. Points in Y correspond to prime ideals containing $\mathfrak{p}$, so the infimum of the dimension of all local rings $\mathcal{O}_{P, X}$ over $P \in Y$ equals to the infimum of the height of all prime ideals of A which contain $\mathfrak{p}$, which is just the height of $\mathfrak{p}$. If Y is any closed subset, then Y corresponds to an ideal $\mathfrak{a}$ in A , and any irreducible closed set of Y contained in Y corresponds to a prime ideal of A that contains $\mathfrak{a}$. From the case when Y is irreducible, the infimum of the codimension of all irreducible closed sets contained in Y equals to the infimum of the height of prime ideals containing $\mathfrak{a}$. The height of such prime ideals is precisely the dimension of the local ring of points $P \in Y$. If X is any integral scheme of finite type over a field k , then we can cover X by a finite number of open affines $U_i = \text{Spec } A_i$ where $\dim A_i = \dim A_j$. If $P \in U_i \cap U_j \cap Y$, then $\dim \mathcal{O}_{P, X} = \dim \mathcal{O}_{P, U_i} = \dim \mathcal{O}_{P, U_j}$; in particular, the prime ideals that P corresponds to in A_i, A_j have the same height. Thus, we can just reduce to the affine case.
- (d) By the same reason as part (c), we can reduce to the affine case, which follows from (§1, 1.7) and (§1, 1.8Ab).
- (e) By (§1, Ex. 1.10b), $\dim U = \sup \dim U \cap U_i = \dim U_i = \dim X$.
- (f) If $U_i = \text{Spec } A_i$ is an open affine cover of X , then $U'_i = U_i \times_k k' = \text{Spec } A_i \otimes_k k'$ is an open affine cover of X' , so by (§1, Ex. 1.10b) and part (a), we reduce to the case when $X = \text{Spec } A$ is affine. Suppose $A = k[x_1, \dots, x_n]/\mathfrak{p}$ for some prime ideal $\mathfrak{p}$. Then X' is the spectra of the ring $k'[x_1, \dots, x_n]/\mathfrak{p}'$, where $\mathfrak{p}'$ is the extension of $\mathfrak{p}$ in $k'[x_1, \dots, x_n]$. The irreducible components of X' correspond to minimal prime ideals of $\mathfrak{p}'$. Let $\mathfrak{q}$ be a minimal prime ideal of $\mathfrak{p}'$ so that $\mathfrak{p} = \mathfrak{q} \cap k[x_1, \dots, x_n]$. We want to show $\dim k[x_1, \dots, x_n]/\mathfrak{p} = \dim k'[x_1, \dots, x_n]/\mathfrak{q}$. The case when k' is an algebraic extension of k is immediate from the going-up going-down theorems, so assume k and k' are algebraically closed. Let K, K' be the fraction fields of $k[x_1, \dots, x_n]/\mathfrak{p}, k'[x_1, \dots, x_n]/\mathfrak{q}$, then by part (b) or (§1, 1.8Aa) it suffices to show

$\text{tr.d.} K/k = \text{tr.d.} K'/k'$. Since k, k' are algebraically closed, K, K' are purely transcendental extensions, so we can write $K = k(y_1, \dots, y_r), K' = k'(y'_1, \dots, y'_{r'})$, where $r = \text{tr.d.} K/k, r' = \text{tr.d.} K'/k'$. By Noether's normalization lemma, we can choose y_i to be linear combinations of $x_1, \dots, x_n$, so after a suitable k -algebra automorphism, we can assume $\mathfrak{p} = (x_1, \dots, x_{n-r})$ so that $y_i = x_i$ for $i = n-r+1, \dots, n$. Then the extension of $\mathfrak{p}$ in $k'[x_1, \dots, x_n]$ is also a prime ideal generated by $x_1, \dots, x_{n-r}$, which implies $r = r'$. Since $n = \dim k[x_1, \dots, x_n] = \dim k'[x_1, \dots, x_n]$, by (§1, 1.8Ab) we have the following corollary:

Corollary 0.4. *Let k'/k be any field extension. If $\mathfrak{p}$ is a prime ideal in $k[x_1, \dots, x_n]$, then any minimal prime ideal of the extension of $\mathfrak{p}$ in $k'[x_1, \dots, x_n]$ has the same height as $\mathfrak{p}$.*

□